

Name: _____

Date: _____

STANDARD FORM, (CALCULATIONS IN STANDARD FORM)

LO: I can add, subtract, multiply and divide numbers in standard form.

Calculating with standard form

Standard form is written as $A \times 10^n$ where $1 \leq A < 10$ and n is an integer. To **multiply or divide**, use index laws on the powers of 10.

Strategy: For multiplication/division, deal with the A values and powers of 10 separately. For addition/subtraction, convert to the same power of 10 first.

Quick examples

●	$(3 \times 10^2) \times (2 \times 10^3) = 6 \times 10^5$	multiply A values, add powers
●	$(8 \times 10^6) \div (4 \times 10^2) = 2 \times 10^4$	divide A values, subtract powers
●	$(4 \times 10^3) + (3 \times 10^3) = 7 \times 10^3$	same power, add A values
●	$(3 \times 10^2) \times (2 \times 10^3) = 6 \times 10^6$	powers add, not multiply

Key idea

Multiply/divide: deal with A and 10^n separately. Add/subtract: match the powers of 10 first.

$$(a \times 10^m) \times (b \times 10^n) = (ab) \times 10^{m+n}$$

Multiply A values, add indices

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - multiplication

Calculate $(4 \times 10^3) \times (2 \times 10^5)$

Multiply A values: $4 \times 2 = 8$

Add powers: $3 + 5 = 8$

Answer: 8×10^8

$$8 \times 10^8$$

Multiply the numbers, add the indices.

Example 2 - division

Calculate $(9 \times 10^7) \div (3 \times 10^3)$

Divide A values: $9 \div 3 = 3$

Subtract powers: $7 - 3 = 4$

Answer: 3×10^4

$$3 \times 10^4$$

Divide the numbers, subtract the indices.

Example 3 - adjusting after multiplication

Calculate $(6 \times 10^4) \times (5 \times 10^3)$

Multiply A values: $6 \times 5 = 30$

Add powers: $4 + 3 = 7$, so 30×10^7

Adjust: $30 = 3 \times 10$, so 3×10^8

$$3 \times 10^8$$

If A ≥ 10 , adjust by increasing the power of 10.

Example 4 - addition

Calculate $(5.2 \times 10^4) + (3.1 \times 10^3)$

Convert: $3.1 \times 10^3 = 0.31 \times 10^4$

Add: $5.2 + 0.31 = 5.51$

Answer: 5.51×10^4

$$5.51 \times 10^4$$

Match the powers of 10 before adding the A values.

YOUR TURN – PRACTICE (deliberate practice)

1. Multiplication

Calculate, giving your answer in standard form.

- a) $(3 \times 10^2) \times (4 \times 10^3)$
- b) $(2 \times 10^5) \times (3 \times 10^4)$
- c) $(5 \times 10^3) \times (4 \times 10^2)$
- d) $(7 \times 10^4) \times (2 \times 10^6)$

2. Division

Calculate, giving your answer in standard form.

- a) $(8 \times 10^6) \div (2 \times 10^3)$
- b) $(9 \times 10^8) \div (3 \times 10^5)$
- c) $(6 \times 10^7) \div (4 \times 10^2)$
- d) $(4.8 \times 10^9) \div (1.2 \times 10^4)$

3. Addition and subtraction

Calculate, giving your answer in standard form.

- a) $(4 \times 10^3) + (5 \times 10^3)$
- b) $(7 \times 10^5) - (2 \times 10^5)$
- c) $(6 \times 10^4) + (8 \times 10^3)$
- d) $(3.5 \times 10^6) - (4 \times 10^5)$

4. Mixed with adjustment

Calculate. Ensure answer is in standard form.

- a) $(8 \times 10^3) \times (5 \times 10^4)$
- b) $(6 \times 10^5) \times (7 \times 10^2)$
- c) $(3 \times 10^{-2}) \times (4 \times 10^5)$
- d) $(2.5 \times 10^4) \times (4 \times 10^{-2})$

5. Worded problems

Answer in standard form.

- a) Light travels 3×10^8 m/s. How far in 500 s?
- b) A virus is 2.5×10^{-7} m. 4000 in a line?
- c) Earth mass: 6×10^{24} kg. Moon: 7.3×10^{22} kg. Difference?
- d) Population 6.7×10^7 shares 2.01 $\times 10^{12}$ equally. Each?
- e) Area = $(4 \times 10^3) \times (2.5 \times 10^2)$. Find area.

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) When multiplying in standard form, add the powers of 10 ☐

Correct answer: _____

Reason: _____

b) $(3 \times 10^2) \times (2 \times 10^3) = 6 \times 10^6$ ☐

Correct answer: _____

Reason: _____

c) If A 10 after calculation, adjust by increasing the power ☐

Correct answer: _____

Reason: _____

d) You can add numbers in standard form by adding the A values regardless of power ☐

Correct answer: _____

Reason: _____

e) To divide in standard form, subtract the powers ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) The distance from Earth to the Sun is approximately 1.5×10^{11} m. Light travels at 3×10^8 m/s. How many seconds does it take light to reach Earth? Give your answer in standard form.

Expression: _____

Simplified: _____

b) Calculate $(4.2 \times 10^5) \times (3 \times 10^{-2}) + (1.74 \times 10^4)$. Show all working and give your answer in standard form.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

